Hand Gesture Recognition with Contour Analysis

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***Abstract*—This mid semester report introduces a vision based hand gesture recognition system based on contour geometry and convexity defects without special sensors. The OpenCV-based model is able to process images, identify contours and extract gestures, which is tested in different lighting and background settings. The report highlights the methodology adopted, experimental design, initial findings, and expected developments to be made in the further stages of development.**

***Index Terms*—Hand Gesture Recognition, Contour Analysis, Convex Hull, Convexity Defects, OpenCV, Computer Vision**

I. Introduction

Hand signals are a naturally interactive way of communication between computers and humans. Gesture recognition is used in sign language recognition, virtual reality, and assistive technology. Although a lot of highly developed models apply sensors or deep learning, they may consume an enormous amount of computation and be costly to carry out.

The current project dwells on a light computer vision-based model based on the evaluation of contours and convexity irregularities to determine gestures in standard camera feeds. The goal is to obtain good recognition in various backgrounds and lighting conditions with a few preprocessing steps and geometric analysis.

II. Literature Review

Computer vision based hand gesture recognition has significantly improved human computer interaction. Rahman et al. [1] created an openCV based system based on contours, convex hulls and convexity defects to identify fingertips and palms effectively, but was unable to perform well in different lighting and complicated backgrounds. This was enhanced by Jaga [2] with HSV-based skin masking and convexity defect angles to recognize in real-time, however, illumination sensitivity remained. Salehin et al. [3] obtained 93.75 percent accuracy in the use of convexity defects although accuracy decreased when there was a cluttered background or when the hands were distant. Fourier Descriptors were found stronger than Hu Moments in Conseil et al. [4], and Fourier Descriptors were improved with Angular Radial Bins (ARB) in Lazarou et al. [5] where Fourier Descriptors used in the rotation- and scale-invariant case enabled real time recognition. In general, the contour-based systems are quick yet sensitive to the environment, whereas shape-based systems are more invariable at a higher computational expense. This work is a blend of both contour geometry and convexity defects with an attempt to increase stability across various light and backgrounds.

III. Methodology

Our approach to hand gesture detection is a rule-based and convexity defect based geometric approach to hand gesture detection. In the HSV and YCrCb color space, we resize, Gaussian blur and skin-color mask hand gesture images. The biggest contour is taken out, and such characteristics as area, convex hull, solidity, centroid, and distances are calculated. Hull angles and distances identify the fingertips whereas convexity defects are used to count the fingers. Gestures are distinguished according to the number of defects and solidity of the contour, fist, one to five. Upon the algorithm run, the predicted label and annotated images of the form of contours, hulls, fingertips, and

Algorithm 1 Hand Gesture Detection using Convexity Defects

Require: Input RGB image I
Ensure: Predicted gesture label G and feature statistics

1. Preprocessing and Extraction

Resize I as needed and apply Gaussian blur with 5x5 kernel

Convert to HSV and YCrCb with their respective threshold as follows:

$$m_1 = \text{inRange}(HSV, H_{low}, H_{high})$$

$$m_2 = \text{inRange}(YCrCb, Y_{low}, Y_{high})$$

Combine: $mask = m_1 \wedge m_2$

Apply opening and closing with kernel K, then Gaussian blur and threshold to get clean mask.

2. Contour Detection

Find contours $\{C_i\}$ in mask.

Select largest contour $C = \arg\max_i \text{area}(C_i)$.

3. Feature Computation

Compute area A, perimeter P, bounding box (w, h):

$$\gamma = \frac{4\pi A}{P^2 + \varepsilon}, \quad \rho = \frac{A}{w \cdot h}$$

Compute convex hull H, hull area A_H, and solidity $s = \frac{A}{A_H}$.

Compute centroid $(c_x, c_y) = \left(\frac{M_{10}}{M_{00}}, \frac{M_{01}}{M_{00}}\right)$.

For each hull vertex p_i, compute distance $d_i = \|p_i - (c_x, c_y)\|$.

4. Fingertip and Convexity Defect Detection

For each p_i with neighbors (p_{i-1}, p_{i+1}):

$$\theta_i = \arccos\left(\frac{(p_{i-1} - p_i) \cdot (p_{i+1} - p_i)}{\|p_{i-1} - p_i\| \|p_{i+1} - p_i\|}\right)$$

If $\theta_i < \theta_{th}$ and $d_i > 0.65\, d_{max}$, mark p_i as fingertip.

Find convexity defects (s, e, f, d) and compute:

$$\alpha = \arccos\left(\frac{b^2 + c^2 - a^2}{2bc}\right), \quad depth = \frac{d}{256}$$

Count defects where $\alpha < \alpha_{th}$ and $depth > 0.1\sqrt{w^2 + h^2}$.

5. Gesture Classification

If $s > 0.9$ and no defects $\Rightarrow$ Fist.

Else if defects = 0 $\Rightarrow$ One, 1 $\Rightarrow$ Two, 2 $\Rightarrow$ Three, 3 $\Rightarrow$ Four, $\geq 4 \Rightarrow$ Five.

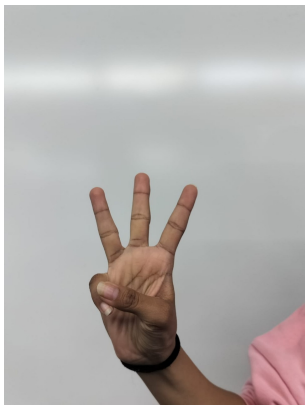
Display contour, hull, fingertips, centroid.

Return G and parameters $(A, s, \rho, \gamma, \text{defect_count})$.

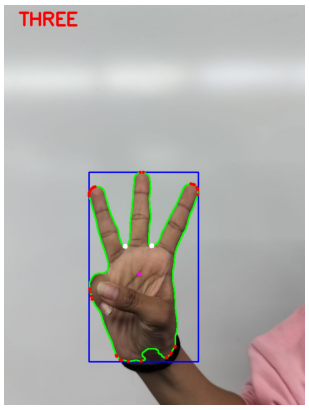
centroid are produced and can be viewed in a visual manner.

IV. Results

The system created could properly recognize hand gestures through contours geometry and convexity defects. The algorithm worked well when being placed under uniform light and plain backdrops. Nevertheless, the performance was worse in dark or crowded conditions, where the partial contours were sometimes used to classify.



(a) Input RGB image



(b) Annotated result: detected hand contour with convex hull and bounding box overlaid

(c) Binary segmentation mask with convexity defects highlighted

Fig. 1: Stages of contour-based gesture processing: detection, hull, and defect visualization.

Green – **Contour lines:** Actual detected hand boundary.
Blue – **Convex hull:** Simplified convex outline of the hand.
Red – **Convexity defects:** Valleys between fingers (used for finger counting).
Red text – **Gesture label:** Output classification result.

V. Discussion

The findings indicate that the use of contour-based geometric techniques is valid to categorize and identify any inert gesture by means of a normal RGB camera. Major challenges include:

- Elasticity to changes in light and colour.
- Inability to sustain the background contours of cluttered backgrounds.
- Less accuracy with hands that are more distant to the camera.

The refinements in the adaptive histogram equalization, automatic ROI tracking, and optimized edge detection will be used to increase stability under the wide range of environments.

VI. Conclusion

An OpenCV based hand gesture recognition system using convex hulls and convexity defects was successfully developed based on a rule-based system. The method is computationally inexpensive and appropriate in real time detection of statical gestures. In my upcoming work, I would concentrate on more powerful preprocessing, more data, and also include dynamic gesture recognition to ensure a constant performance in real-life situation.

References

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